DUNMAN SECONDARY SCHOOL

CANDIDATE NAME			
CLASS		**INDEX NUMBER**	

PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS

MATHEMATICS **4052/01**

Paper 1

22 August 2023

2 hours 15 minutes

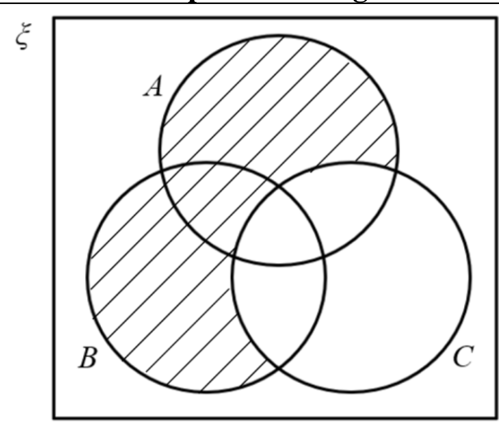
Marking Scheme

(Updated 1 Dec 2022 by HOD/Math aligned to Marking Scheme for Specimen Papers 4049)

Question	Answer
1	$\frac{2}{3}$

Question	Answer
2	$(kx+3)(x+2)=0$
	$x=-2$ or $-\frac{3}{k}$

Question	Answer
3	$3.4 \div 5.3 = 0.64150$
	$0.64150 = 6.4150 \times 10^{-1}$
	$p = m - n - 1$

Question	Answer
4a	It is the set of **non-prime[1] integers[2] less than 10 [3]**. OR It is the set of **positive integers[1] less than 10[2]** and are **not prime[3].**
4b	

Question	Answer
5	$F = \frac{k}{d^2}$
	When $d = a,\ F_1 = \frac{k}{a^2}$
	When $d = 0.4a,\ F_2 = \frac{k}{(0.4a)^2}$
	$= \frac{k}{0.16a^2}$
	Percentage increase $= \frac{\frac{k}{0.16a^2} - \frac{k}{a^2}}{\frac{k}{a^2}} \times 100\% = \frac{\frac{1}{0.16} - 1}{1} \times 100\%$
	$= 525\%$

Question	Answer
6	$14 = \frac{1}{2}(7)(5)\sin\angle ABC$
	$\sin\angle ABC = \frac{14\times 2}{(7)(5)}$
	$\angle ABC = \sin^{-1}\left(\frac{14\times 2}{(7)(5)}\right)$
	$\angle ABC = 53.1°$ or $180 - 53.1°$
	$\angle ABC = 126.9°$

Question	Answer
7	One number has to be 11 since mode is 11.
	One number has to be 13 since range is 10.
	The remaining number has to be 8 since median is 8
	$x + y + z = 32$

Question	Answer
8a	The range for the vertical **axis** is too large for the data represented.
8b	The reader may think the rise in temperature is insignificant.

Question	Answer
9	$\frac{3}{2-x}+\frac{x}{x^2-3x+2}$
	$=-\frac{3}{x-2}+\frac{x}{(x-2)(x-1)}$
	$=-\frac{3(x-1)}{(x-2)(x-1)}+\frac{x}{(x-2)(x-1)}$
	$=\frac{3-2x}{(x-2)(x-1)}$

Question	Answer
10a	$\angle CBD = 4x+12$
	OR
	$\angle CBD = 90-2x$
10b	$4x+12=90-2x$ OR $2x+4x+12=90$
	$x=13$
10c	Any point clearly marked with **x** or ● and labelled on the minor arc between *B* and *D*.

Question	Answer
11a	$(x+3)^2-20$
11b	$(x+3)^2-20=0$
	$x+3=\pm\sqrt{20}$
	$x=1.47$ or -7.47

Question	Answer
12	For Los Angeles, USD7.90 = SGD10.586
	1 gallon = 3.785 litres
	1 litre costs $\frac{10.586}{3.785} = \text{SGD}2.7968$
	Cheaper in LA as SGD2.80 for 1 litre in LA, but SGD2.88 in SG.
	OR
	For Singapore, SGD2.88 = USD2.1492
	1 litre = 0.26420 gallon
	1 gallon costs $\frac{2.1492}{0.26420} = \text{USD}8.1349$
	Cheaper in LA as USD8.13 for 1 gallon in LA, but USD7.90 in SG.

Question	Answer
13a	$gradient = \dfrac{\text{change in } F}{\text{change in } C} = \dfrac{9}{5}$
	$27 = \dfrac{9}{5}C$
	$C = 15$
	OR
	$F_1 = \dfrac{9}{5}C_1 + 32$
	$F_2 = \dfrac{9}{5}C_2 + 32$
	$F_1 - F_1 = 27$
	$\dfrac{9}{5}C_1 - \dfrac{9}{5}C_2 = 27$
	$C_1 - C_2 = 27 \times \dfrac{5}{9}$
	difference = 15
13b	No, because when C is 0, F is not zero.
	OR
	No, because F and C are not related by $F = kC$.

Question	Answer
14	$10000 + 3159.31 = 10000(1 + \dfrac{r}{100})^7$
	$(1 + \dfrac{r}{100})^7 = \dfrac{13159.31}{10000}$
	$1 + \dfrac{r}{100} = \sqrt[7]{\dfrac{13159.31}{10000}}$
	$1 + \dfrac{r}{100} = 1.0400$
	$r = 4$ OR 4.00

Question	Answer
15	(11, 11)

Question	Answer
16	From the turning point, $y=-k(x-1)^2+8$
	Substituting (0, 6), $6=-k(0-1)^2+8$
	$k=2$
	$y=-2(x-1)^2+8$
	$y=-2x^2+4x+6$
	OR
	From the turning point, $y=-k(x-1)^2+8$

Question	Answer
17a	Using (0, –3), $-3=-ka^0$
	$k=3$
	Using (2, –8), $-12=-3a^2$
	$a=2$
17b	Distance = $\sqrt{[-3-(-12)]^2+(0-2)^2}$
	= 9.22 units

Question	Answer
18a	$2^2\times3^2\times7$
18b	7
18c	270

Question	Answer
19a	$\cos(1) = \dfrac{5}{AC}$
	$AC = \dfrac{5}{\cos(1)}$
	$CD = \dfrac{5}{\cos(1)} - 5$
	$CD = 4.2540 = 4.25$ cm
19b	Arc $BD = 5 \times 1$
	Arc $DE = 4.2540 \times (\frac{\pi}{2} - 1)$
	$\tan(1) = \dfrac{EC}{5}$
	$EC = 5 \tan(1)$
	$BE = 5 \tan(1) - 4.2540$
	Perimeter $= 5 \times 1 + 4.2540 \times (\frac{\pi}{2} - 1) + [5 \tan(1) - 4.2540]$
	$= 11.0$ (10.961)

Question	Answer
20a	930
20b	1 cm : 50 km
	1 cm^2 : 2500 km^2
	$\dfrac{5765}{2500} = 2.306\ cm^2$

Question	Answer
21a	96
21b	45
21c	136 students has less than 70 = 24 students has more than 70
	$\dfrac{3}{20}$ or 0.15

Question	Answer
22	$a = \dfrac{kr^2}{r^2 - x}$
	$ar^2 - ax = kr^2$
	$ar^2 - kr^2 = ax$
	$r^2(a-k) = ax$
	$r = \pm\sqrt{\dfrac{ax}{a-k}}$

Question	Answer
23a	$\left\lvert\overrightarrow{AB}\right\rvert = \sqrt{7^2 + (-3)^2}$
	$= 7.62$ unit
23b	Gradient = $\dfrac{-3}{7}$
	$y-(-2) = -\dfrac{3}{7}(x-3)$
	$y = -\dfrac{3}{7}x - \dfrac{5}{7}$
23c	$\overrightarrow{CD} = \begin{pmatrix} 11 \\ 8 \end{pmatrix} - \begin{pmatrix} -3 \\ 14 \end{pmatrix}$
	$\overrightarrow{CD} = \begin{pmatrix} 14 \\ -6 \end{pmatrix}$
	$\overrightarrow{CD} = 2\begin{pmatrix} 7 \\ -3 \end{pmatrix}$
	Since $\overrightarrow{CD} = 2\overrightarrow{AB}$, AB is parallel to CD.

Question	Answer
24a	Let the number of additional volunteers be x
	$\dfrac{23+x}{23+x+17} > 70\%$
	$\dfrac{23+x}{40+x} > 0.7$
	$23+x > 0.7(40+x)$
	$23+x > 28+0.7x$
	$0.3x > 5$
	$x > 16\dfrac{2}{3}$
	$\therefore 17$
24b	$0.8\times 1.2 = 0.96$
	Yes, it made a loss of 4%.

Question	Answer
25a	$\begin{pmatrix} 80 & 130 & 170 \\ 120 & 180 & 230 \end{pmatrix}$
25b	$\begin{pmatrix} 80\times 60+130\times 40+170\times 15 \\ 120\times 60+180\times 40+230\times 15 \end{pmatrix}$
	$=\begin{pmatrix} 12550 \\ 17850 \end{pmatrix}$
25c	12550 and 17850 represents the total costs of booking all the rooms on a weekday and a weekend respectively.
25d	$\begin{pmatrix} 3 & 0 \end{pmatrix}$

Question	Answer
26a	$\frac{5}{14}\times\frac{4}{13}$
	$=\frac{10}{91}$
26b	$\frac{7}{14}\times\frac{2}{14}+\frac{5}{14}\times\frac{2}{13}+\frac{2}{14}\times\frac{1}{14}$
	$=\frac{87}{637}$

Question	Answer
27	By the property of equal chords, $\cos 20° = \frac{5}{r}$
	$r = \frac{5}{\cos 20°} = 5.3208$ cm
	Let the centre of the circle be O,
	$\angle QOR = 180° - 2\times 20° = 140°$
	Area of sector QOR = $\frac{140°}{360°}\times\pi\times(5.3208)^2$
	Area of triangle QOR = $\frac{1}{2}\times(5.3208)^2\times\sin(140°)$
	Area of shaded region = 25.5 cm^2